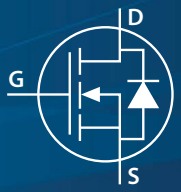
EPC2306 – Enhancement Mode Power Transistor

V_{DS} , 100 V
$R_{DS(on)}$, 3.8 mΩ max

PRELIMINARY



EPC EFFICIENT POWER CONVERSION

RoHS Halogen-Free

Gallium Nitride's exceptionally high electron mobility and low temperature coefficient allows very low $R_{DS(on)}$, while its lateral device structure and majority carrier diode provide exceptionally low Q_G and zero Q_{RR}. The end result is a device that can handle tasks where very high switching frequency, and low on-time are beneficial as well as those where on-state losses dominate.

Application Notes:

- Easy-to-use and reliable gate, Gate Drive ON = 5 V typical, OFF = 0 V (negative voltage not needed)
- Top of FET is electrically connected to source

Questions:

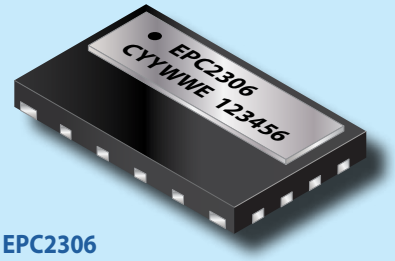
ASK AN EXPERT GaN

Maximum Ratings			
PARAMETER		**VALUE**	**UNIT**
V_{DS}	Drain-to-Source Voltage (Continuous)	100	V
	Drain-to-Source Voltage (up to 10,000 5 ms pulses at 150°C)	120	
I_D	Continuous (T_A = 25°C)	48	A
	Pulsed (25°C, T_{PULSE} = 300 µs)	197	
V_{GS}	Gate-to-Source Voltage	6	V
	Gate-to-Source Voltage	-4	
T_J	Operating Temperature	-40 to 150	°C
T_{STG}	Storage Temperature	-40 to 150	

Thermal Characteristics			
PARAMETER		**TYP**	**UNIT**
$R_{\theta JC}$	Thermal Resistance, Junction-to-Case (Case TOP)	0.5	°C/W
$R_{\theta JB}$	Thermal Resistance, Junction-to-Board (Case BOTTOM)	3.0	
$R_{\theta JA_JEDEC}$	Thermal Resistance, Junction-to-Ambient (using JEDEC 51-2 PCB)	54	
$R_{\theta JA_EVB}$	Thermal Resistance, Junction-to-Ambient (using EPC90145 EVB)	23	

Static Characteristics (T_J = 25°C unless otherwise stated)						
PARAMETER		**TEST CONDITIONS**	**MIN**	**TYP**	**MAX**	**UNIT**
BV_{DSS}	Drain-to-Source Voltage	V_{GS} = 0 V, I_D = [TBD]	100			V
I_{DSS}	Drain-Source Leakage	V_{GS} = 0 V, V_{DS} = 80 V		0.5		µA
I_{GSS}	Gate-to-Source Forward Leakage	V_{GS} = 5 V		0.005	1.9	mA
	Gate-to-Source Forward Leakage#	V_{GS} = 5 V, T_J = 125°C		0.2	4.2	
	Gate-to-Source Reverse Leakage	V_{GS} = -4 V		0.005		
$V_{GS(TH)}$	Gate Threshold Voltage	V_{DS} = V_{GS}, I_D = 7 mA	0.8	1.3	2.5	V
$R_{DS(on)}$	Drain-Source On Resistance	V_{GS} = 5 V, I_D = 25 A		3.0	3.8	mΩ
V_{SD}	Source-Drain Forward Voltage#	I_S = 0.5 A, V_{GS} = 0 V		1.6		V

Defined by design. Not subject to production test.



EPC2306
Package size: 3 x 5 mm

Applications

- AC-DC chargers, SMPS, adaptors, power supplies
- High Frequency DC-DC Conversion up to 80 V input (Buck, Boost, Buck-Boost and LLC)
- 24 V–60 V Motor Drives
- High Power Density DC-DC modules from 40 V– 60 V to 5 V–12 V
- Synchronous Rectification
- Solar MPPT

Benefits

- Higher Efficiency – Lower conduction and switching losses, zero reverse recovery losses
- Ultra Small Footprint – Higher power density

Scan QR code or click link below for more information including reliability reports, device models, demo boards!

https://l.ead.me/EPC2306

Dynamic Characteristics# (T_J = 25°C unless otherwise stated)						
PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
C_{ISS}	Input Capacitance	V_{DS} = 50 V, V_{GS} = 0 V		1667	2366	pF
C_{RSS}	Reverse Transfer Capacitance			3.6		
C_{OSS}	Output Capacitance			482	559	
$C_{OSS(ER)}$	Effective Output Capacitance, Energy Related (Note 1)	V_{DS} = 0 to 50 V, V_{GS} = 0 V		593		
$C_{OSS(TR)}$	Effective Output Capacitance, Time Related (Note 2)			767		
R_G	Gate Resistance			0.4		Ω
Q_G	Total Gate Charge	V_{DS} = 50 V, V_{GS} = 5 V, I_D = 25 A		11.6	16.3	nC
Q_{GS}	Gate to Source Charge	V_{DS} = 50 V, I_D = 25 A		4.1		
Q_{GD}	Gate-to-Drain Charge			0.8		
$Q_{G(TH)}$	Gate Charge at Threshold			3		
Q_{OSS}	Output Charge	V_{DS} = 50 V, V_{GS} = 0 V		38	47	
Q_{RR}	Source-Drain Recovery Charge			0		

All measurements were done with substrate shorted to source.
Defined by design. Not subject to production test.
Note 1: $C_{OSS(ER)}$ is a fixed capacitance that gives the same stored energy as C_{OSS} while V_{DS} is rising from 0 to 50% BV_{DSS}.
Note 2: $C_{OSS(TR)}$ is a fixed capacitance that gives the same charging time as C_{OSS} while V_{DS} is rising from 0 to 50% BV_{DSS}.

Figure 1: Typical Output Characteristics at 25°C

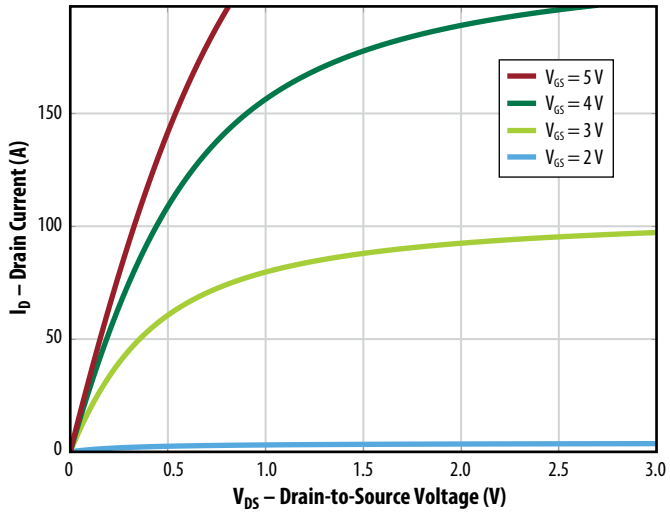


Figure 2: Typical Transfer Characteristics

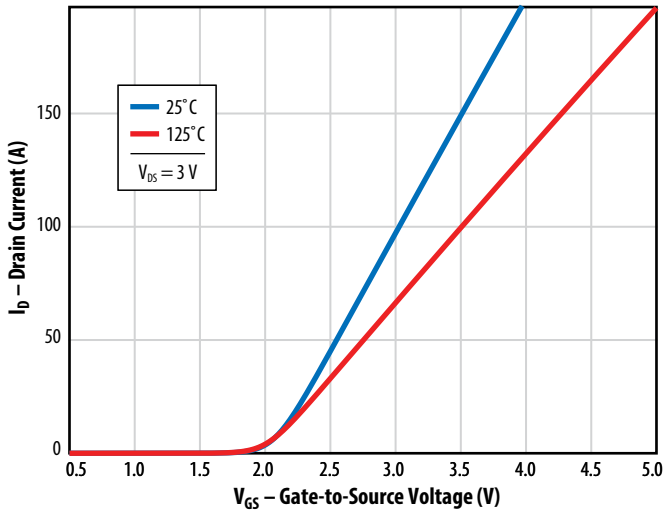


Figure 3: Typical $R_{DS(on)}$ vs. V_{GS} for Various Drain Currents

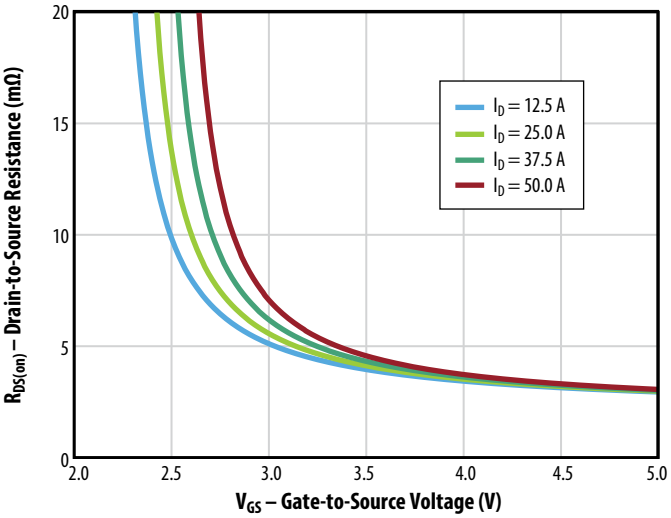


Figure 4: Typical $R_{DS(on)}$ vs. V_{GS} for Various Temperatures

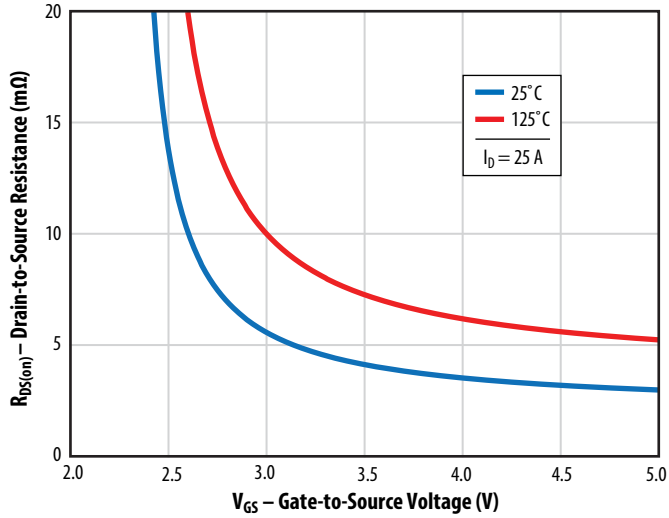


Figure 5a: Typical Capacitance (Linear Scale)

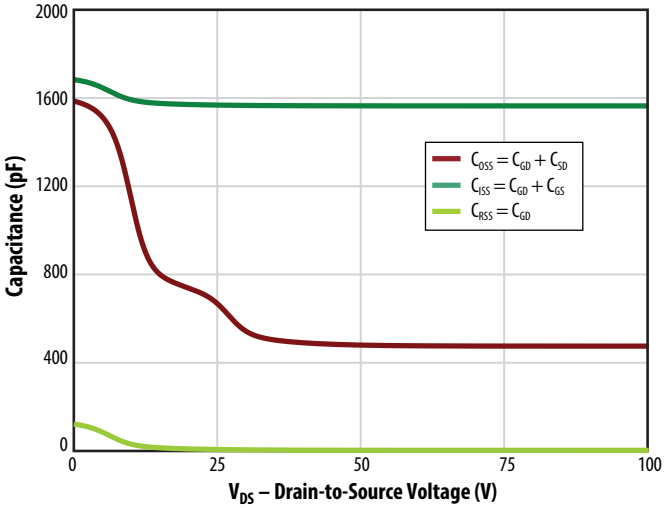


Figure 5b: Typical Capacitance (Log Scale)

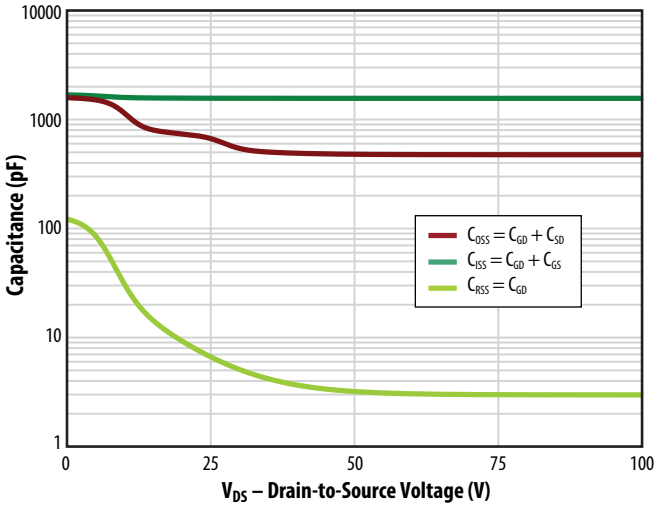


Figure 6: Typical Output Charge and C_{OSS} Stored Energy

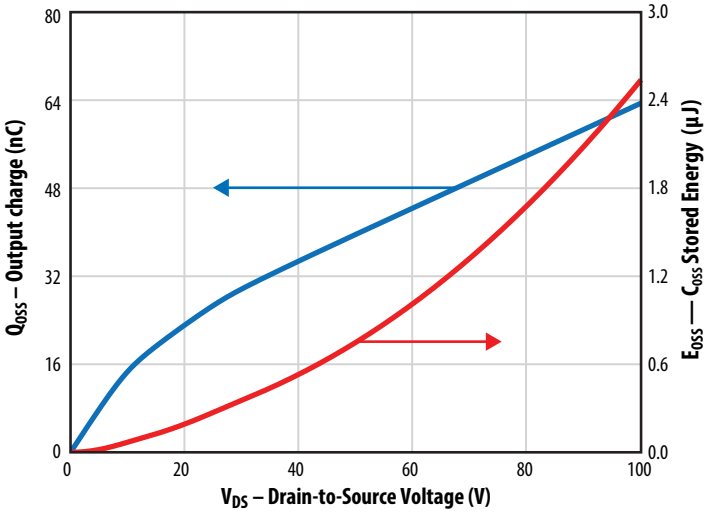


Figure 7: Typical Gate Charge

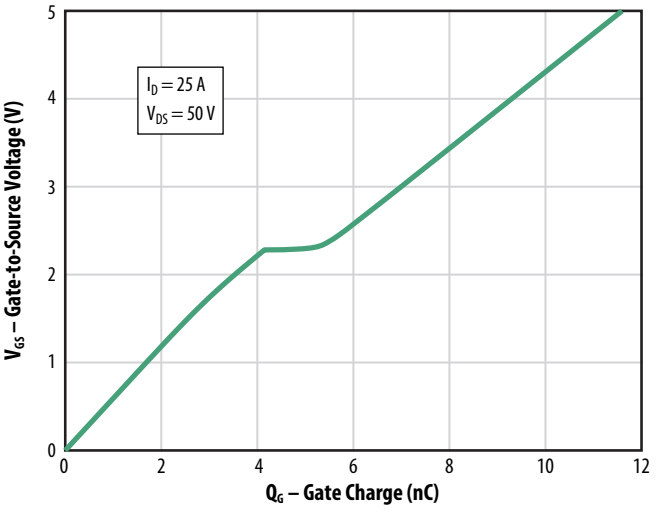
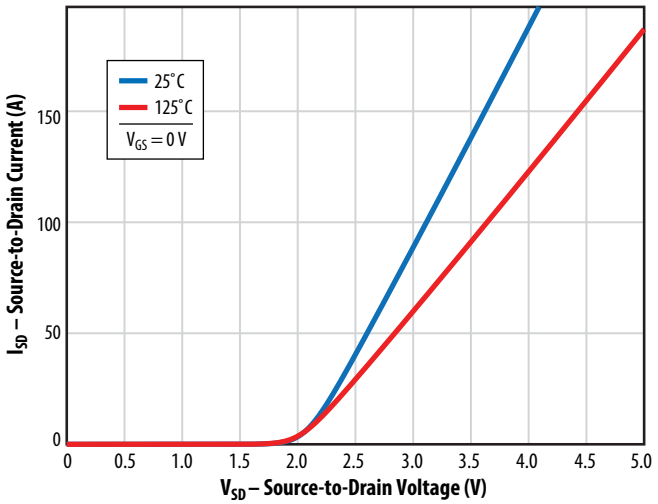


Figure 8: Typical Reverse Drain-Source Characteristics



Note: Negative gate drive voltage increases the reverse drain-source voltage. EPC recommends 0 V for OFF.

Figure 9: Typical Normalized On-State Resistance vs. Temp.

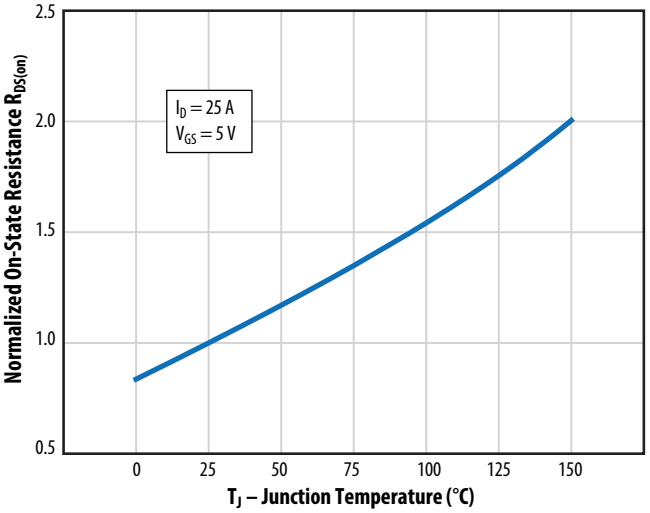


Figure 10: Typical Normalized Threshold Voltage vs. Temp.

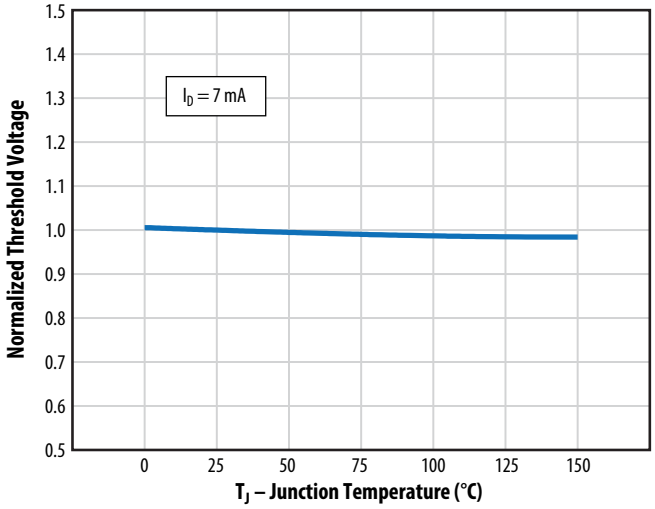


Figure 11: Safe Operating Area

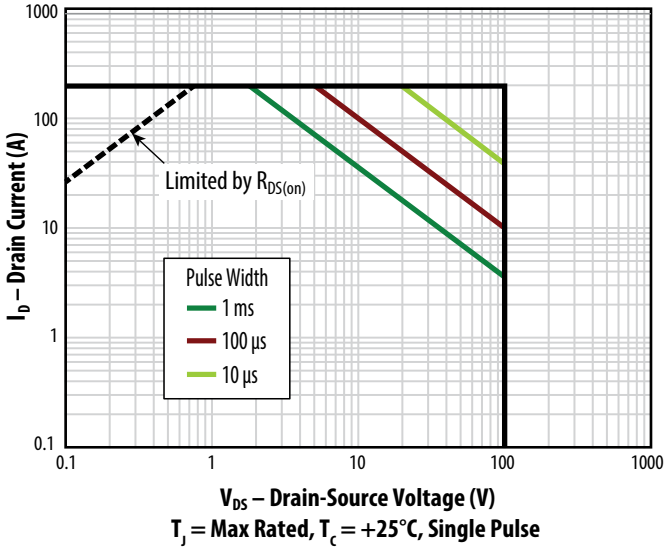
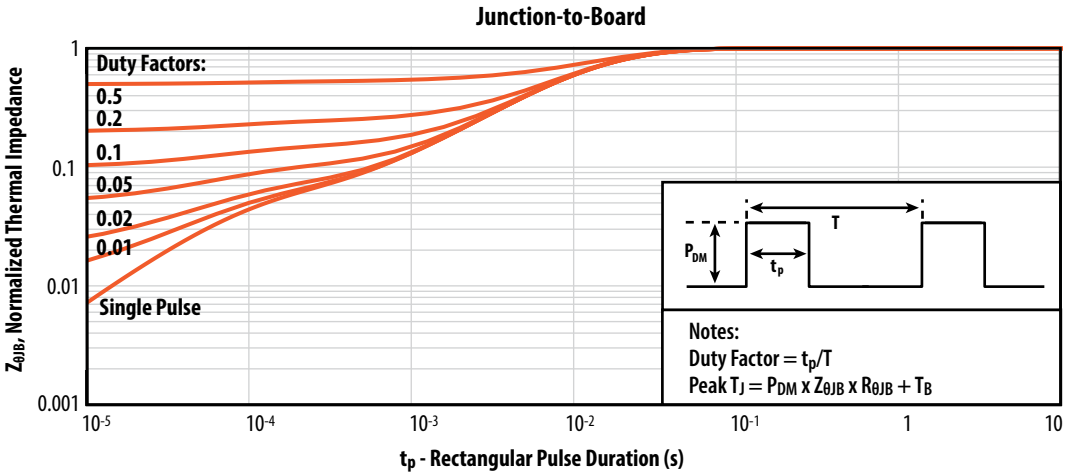
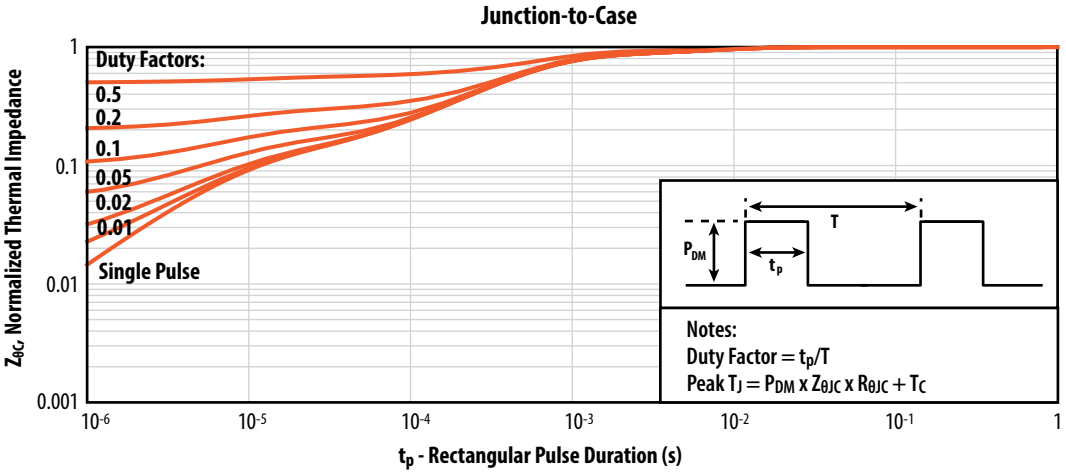


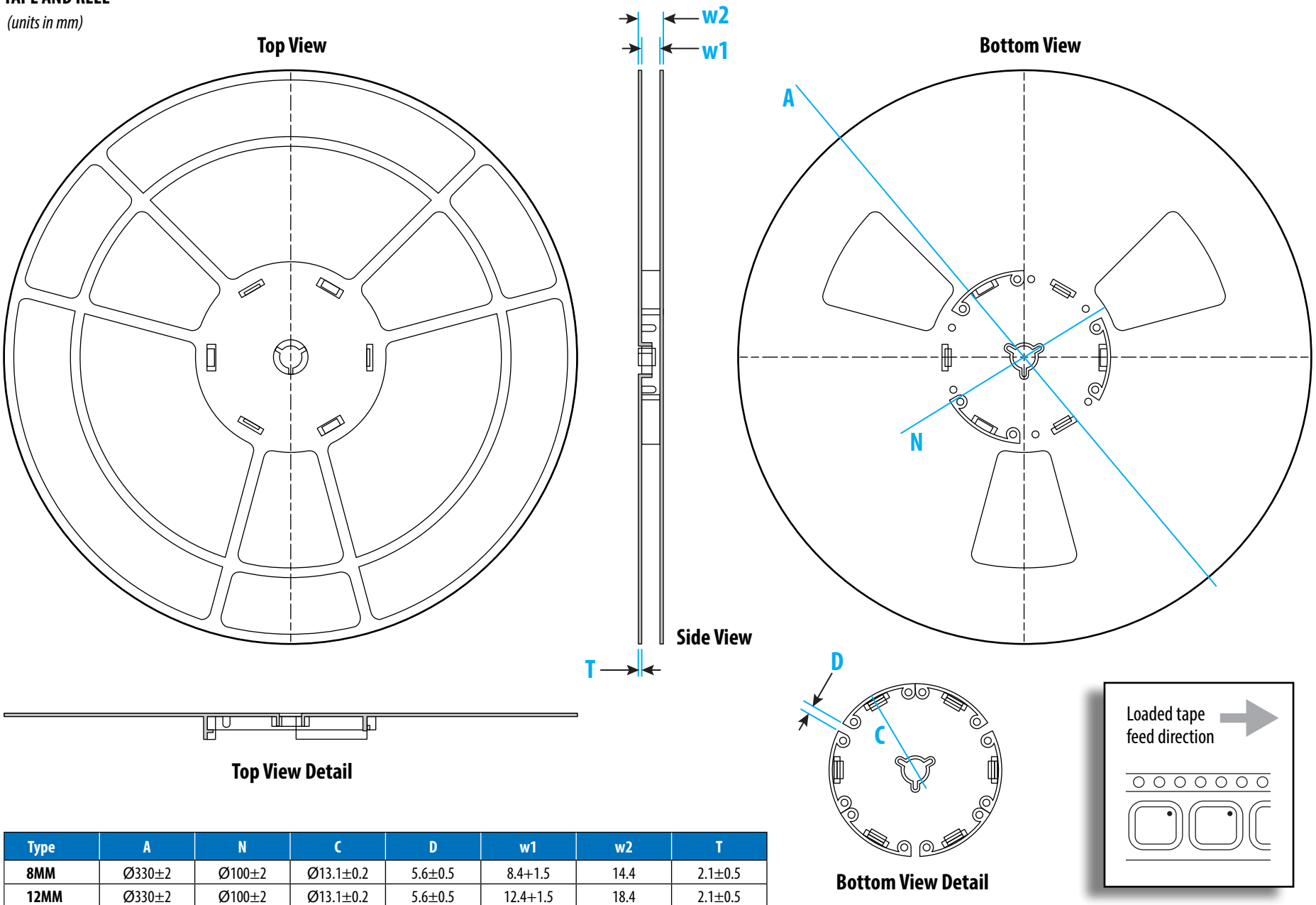
Figure 12: Transient Thermal Response Curves



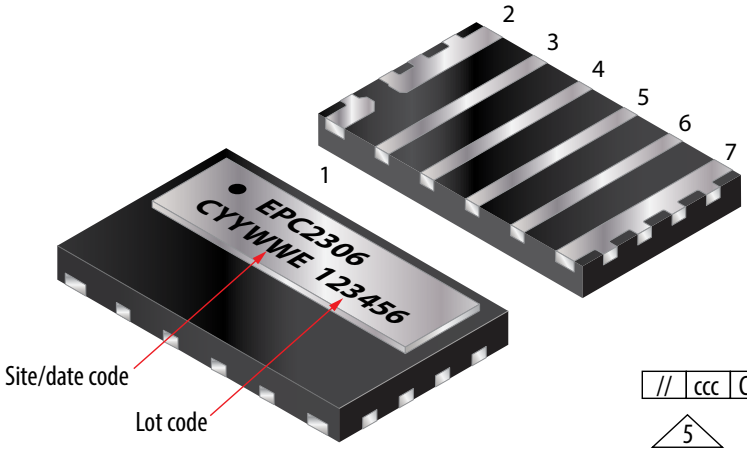


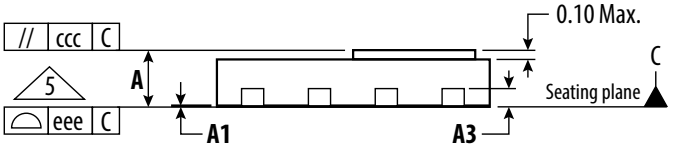
TAPE AND REEL

(units in mm)

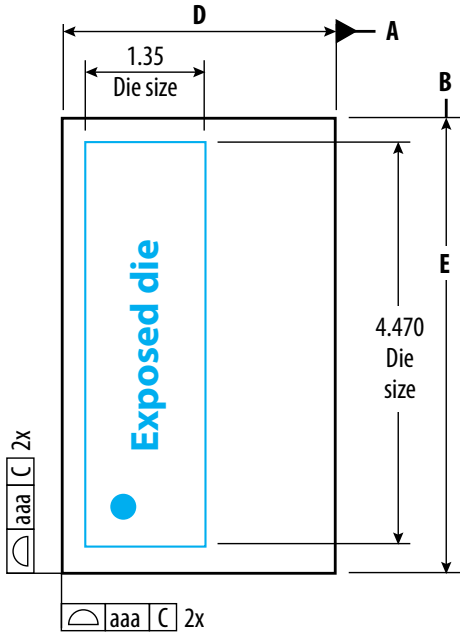


Type	A	N	C	D	w1	w2	T
8MM	Ø330±2	Ø100±2	Ø13.1±0.2	5.6±0.5	8.4+1.5	14.4	2.1±0.5
12MM	Ø330±2	Ø100±2	Ø13.1±0.2	5.6±0.5	12.4+1.5	18.4	2.1±0.5
16MM	Ø330±2	Ø100±2	Ø13.1±0.2	5.6±0.5	16.4+1.5	22.4	2.1±0.5
24MM	Ø330±2	Ø100±2	Ø13.1±0.2	5.6±0.5	24.4+1.5	30,4	2.1±0.5

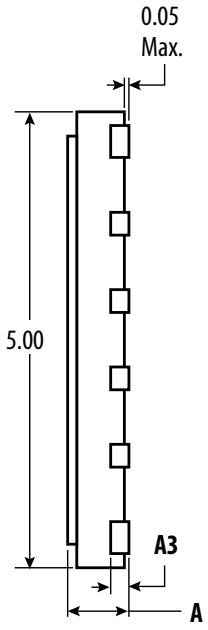




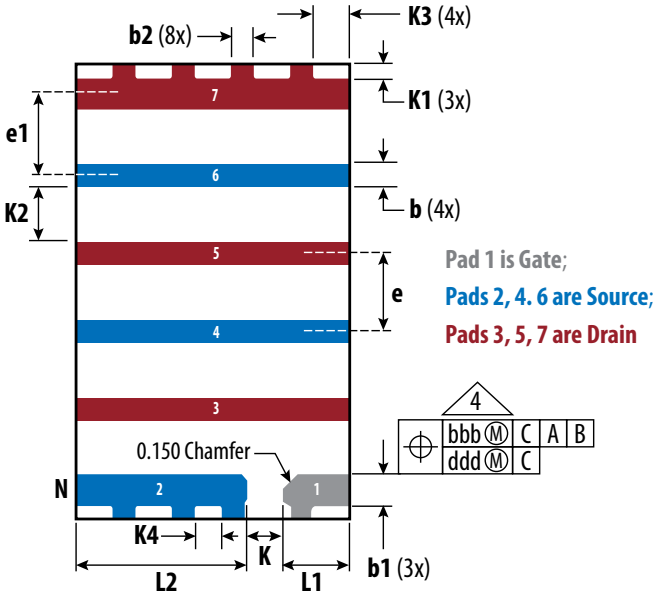
Side View 2



Top View



Side View 1



Bottom View

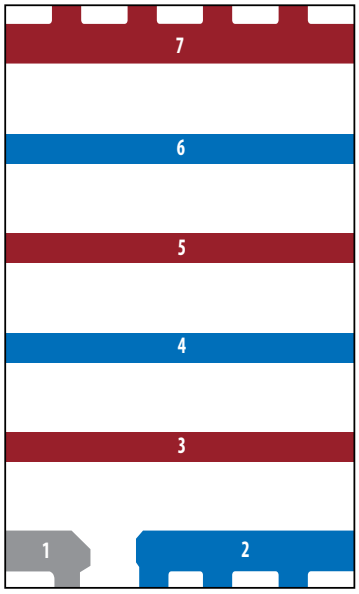
SYMBOL	Dimension (mm)			
	MIN	Nominal	MAX	Note
A	0.60	0.65	0.70	
A1	0.00	0.02	0.05	
A3		0.20 Ref		
b	0.20	0.25	0.30	4
b1	0.30	0.35	0.40	4
b2	0.20	0.25	0.30	4
D		3.00 BSC		
E		5.00 BSC		
e		0.85 BSC		
e1		0.90 BSC		
L1	0.625	0.725	0.825	
L2	1.775	1.875	1.975	

SYMBOL	Dimension (mm)			
	MIN	Nominal	MAX	Note
K	0.35	0.40	0.45	
K1	0.10	0.15	0.20	
K2	0.55	0.60	0.65	
K3	0.35	0.40	0.45	
K4	0.25	0.30	0.35	
aaa		0.05		
bbb		0.10		
ccc		0.10		
ddd		0.05		
eee		0.08		
N		15		3
NE		6		

Notes:

1. Dimensioning and tolerancing conform to ASME Y14.5-2009
2. All dimensions are in millimeters
3. **N** is the total number of terminals
4. Dimension **b** applies to the metallized terminal. If the terminal has a radius on the other end of it, dimension **b** should not be measured in that radius area.
5. Coplanarity applies to the terminals and all the other bottom surface metallization.

TRANSPARENT VIEW

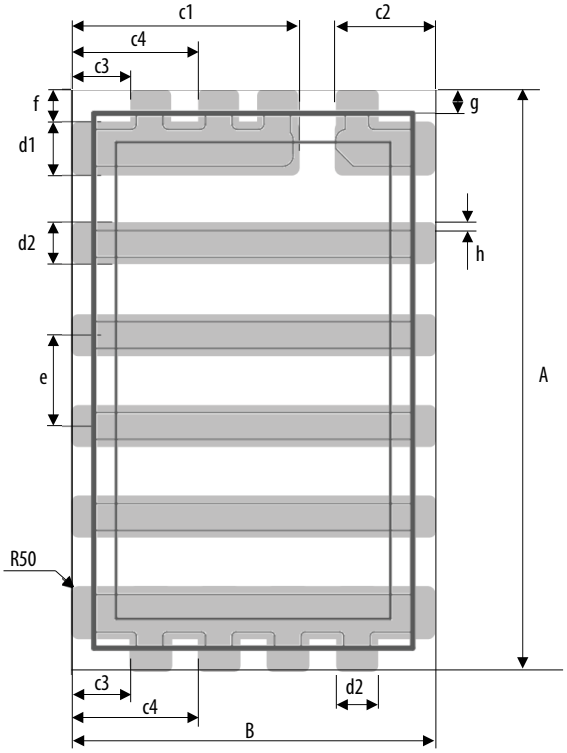


Transparent Top View

PIN	DESCRIPTION
1	Gate
2	Source
3	Drain
4	Source
5	Drain
6	Source
7	Drain

RECOMMENDED LAND PATTERN

(units in mm)



Land pattern is solder mask defined.

It is recommended to have on-Cu trace PCB vias.

DIM	Nominal
A	5.4
B	3.4
c1	2.1
c2	0.91
c3	0.54
c4	1.19
c5	0.985
d1	0.47
d2	0.37
e	0.85
f	0.29
g	0.2
h	0.06

Additional resources available:

- Assembly resources – https://epc-co.com/epc/Portals/0/epc/documents/product-training/Appnote_GaNassembly.pdf
- Library of Altium footprints for production FETs and ICs – https://epc-co.com/epc/documents/altium-files/EPC%20Altium%20Library.zip (for preliminary device Altium footprints, contact EPC)

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